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May 2, 2026

Editor-in-Chief
IEEE Transactions on Medical Imaging

Dear Editor-in-Chief,

I am submitting the manuscript **"A 16-Bit-Native Entropy Coding Pipeline for Lossless Medical Image Compression"** for consideration as a Full Paper in *IEEE Transactions on Medical Imaging*.

Summary. This paper presents MIC, a lossless compression codec for high-bit-depth Digital Imaging and Communications in Medicine (DICOM) images. The central contribution is a pipeline that performs spatial prediction, 16-bit-native run-length encoding, and large-alphabet Finite State Entropy (FSE) coding directly on 16-bit residuals, rather than splitting pixels into byte pairs before entropy coding. I show theoretically and empirically that byte-splitting incurs a structural information loss of 0.3–1.1 bits per symbol on delta-coded medical residuals (8–22% of total entropy), and that a 16-bit-native coder recovers this cost. The paper additionally contributes a four-state interleaved FSE decoder that improves decompression throughput by a geometric mean of 68% on ARM64 and 43% on AMD64 with no change to compressed size, a parallel strip (PICS) mode that exceeds High-Throughput JPEG 2000 (HTJ2K) decompression speed on all 21 tested images on both platforms, and a approximately 20-kilobyte pure JavaScript decoder enabling client-side in-browser DICOM viewing without server-side transcoding.

Relevance to IEEE TMI. The manuscript addresses lossless compression for clinical DICOM data—a core challenge in medical image storage, PACS transmission, and diagnostic viewing of 10–16 bit pixel data. Results are reported on 21 de-identified clinical DICOM images spanning nine modalities (CT, MRI, mammography, computed radiography, radiography, fluoroscopy, nuclear medicine, secondary capture, X-ray), with fair in-process comparison against HTJ2K (OpenJPH) and JPEG-LS (CharLS). The combination of compression-ratio analysis, throughput benchmarking, and browser deployment feasibility is directly relevant to the TMI readership working on medical image informatics systems.

Originality. This work has not been submitted to, or published in, any other journal. The manuscript represents a complete original work; no prior conference or workshop version has been published.

Open-source reproducibility. The complete implementation, benchmark harness, and test data references are available at `https://github.com/pappuks/medical-image-codec`. All results are reproducible using the commands documented in the repository.

Ethics. The evaluation dataset consists entirely of publicly available, de-identified DICOM images from the NEMA WG-04 compsamples archive, the NEMA 1997 CD, and the Clunie Breast Tomosynthesis public case. No human subjects research was conducted; no institutional ethics approval is required per DICOM PS 3.15 Appendix E.

Conflict of interest. The author declares no competing financial interests or personal relationships that could have influenced this work.

Suggested Associate Editor area. Medical Image Compression; Medical Image Storage and Transmission; Medical Imaging Informatics Systems.

Note on suggested reviewers. I note that the paper compares against OpenJPH and CharLS; authors of those libraries may have a conflict of interest and could reasonably be excluded at the Editor's discretion.

I confirm this is a single-author submission and that the manuscript has been prepared in accordance with IEEE Transactions on Medical Imaging author guidelines, including the 10-page limit, the 250-word abstract limit, and the use of de-identified public datasets.

Sincerely,

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https://github.com/pappuks/medical-image-codec